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Relationship between aggregation of therapeutic proteins and parameters: Acceleration and frequency



agitation

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ABSTRACT

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An increase in protein aggregates during transportation should be suppressed in therapeutic protein products because the aggregates have a potential risk of immunogenicity. In this study, three protein solutions in vials were exposed to tri-axial vibration with various combinations of frequency and acceleration using a transportation test system to investigate the relationship between low g-force stresses and protein aggregate generation. The number concentration of micron aggregates detected by flow

Introduction

Therapeutic proteins, including monoclonal antibodies and Fc fusion proteins, are important classes of biopharmaceuticals, the use of which is growing rapidly.¹ Antibodies, a major class of biotherapeutics, have low adverse effects partly because of their high specificity of binding to target molecules.² A drawback of therapeutic proteins is their inherent instability, resulting in generating aggregates, because the aggregates have a potential risk of immunogenicity.^{3,4,5} Although the relationship between immunogenicity and protein aggregates has been widely investigated, this relationship remains incompletely understood. Some studies showed that nanometer aggregates (<100 nm)⁶ and silicone oil droplets with monoclonal antibody⁷ did not increase immunogenicity,

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aggregates,⁸ submicrometer aggregates (100–1000 nm),⁹ and micron aggregates.¹⁰ Considering that these conflicting conclusions among previous papers, it is important to evaluate and minimize formation of the whole range of aggregates to avoid unwanted immunogenicity. Although submicrometer, micron and larger aggregates are often referred to as particles, the word aggregates are used in this study in line with the nomenclature.¹¹ For the same reason, the word nanometer aggregates are used for the nanometer-sized aggregates and oligomers of the proteins.

imaging analysis increased markedly when the acceleration and frequency of agitation were within a specific range, in other words, above a threshold. This threshold was common among the three protein solutions. The suppression of micron aggregate formation by adding a surfactant suggested that agitation above the threshold increased micron aggregates mainly via interface-mediated routes. Notably, agitation, including agitation below the threshold, accelerated spontaneous oligomerization (nanometer aggregate generation) of proteins in bulk solution even in the presence of the surfactant. Studies of stability against mechanical stresses (e.g., a random vibration test to simulate actual shipment, with a time-compressed setting by increasing acceleration) need to be performed and discussed with careful consideration of the threshold for generating micron aggregates.

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whereas other reports showed high immunogenicity of nanometer

Proteins are exposed to various mechanical stresses during manufacturing, filling, storage, transportation, and injection.^{12,13,14} These mechanical stresses have been

recognized to induce interfacemediated aggregate generation (e.g., solid-liquid, air-liquid, and oil-liquid interfaces).^{15,16,17} Proteins adsorb to interfaces because of their partial amphiphilic nature.¹⁸ Adsorption of proteins to interfaces leads to the formation of protein films.^{19,20}

Protein films can be eroded and become protein aggregates by mechanical stresses such as flow,^{21,22} abrasion,²³ cavitation,²⁴ and cycles of compression/dilation.^{16,25,26} As these mechanical stresses are inevitable, it is essential to optimize protein sequences, formulations, and packaging to improve stability against mechanical stresses.

Protein solutions are subjected to two types of mechanical stress during transportation: low-g-force mechanical stresses such as truck vibrations and high-g-force mechanical shocks like dropping. Over the last few years, the identification of dominant stresses that increase protein aggregates during transportation has been discussed. Previously high-g-force stresses were regarded as the primary cause of protein aggregates during

transportation.^{21,24} Based on this assumption, previous studies evaluated the impact of high g-force shock events and showed that aggregate generation can be mitigated through improved packaging.^{27,28,29} In contrast, the importance of low-g-force stresses during transportation has only recently been recognized. Witeof et al. reported that micron aggregates generated by low-g-force stresses produced by an orbital shaker at 320 rpm, rather than high-g-force shock events generated by multiple dropping, showed higher morphological similarity to micron aggregates reported in actual shipping.²⁷ They suggested that low-g-force stresses dominate aggregate generation during transportation. Another study demonstrated that aggregates generated by only high g-force stress did not match the data from actual shipping.²⁹ Moreover, a random vibration test including low-g-force stresses with acceleration of less than 6 G, in accordance with a previous version of the American Society for Testing and Materials standards (ASTM) D4169,³⁰ successfully simulated micron aggregate behavior during actual shipment for several monoclonal antibodies in vials and prefilled syringe configurations.³¹

Low-g-force stresses such as vibration during transportation by air or truck are repetitive, and thus the frequency and acceleration of the vibration are characteristic parameters. Clarifying the relationship between these parameters and protein aggregate generation is important when establishing an appropriate shipping study and therefore essential for optimizing protein sequences, formulations, and packaging. Johann et al. have recently examined critical parameters for a microplate-based orbital shaker method to screen optimal protein sequences and formulations.³² This study compared trends in monomer recovery as a function of shaking frequency from 0 to 3000 rpm and liquid height, revealing that as the air-liquid interfacial area increased because of increasing frequency, the monomer content significantly decreased. In addition, Fleischman et al. compared micron aggregates of monoclonal antibodies after stress with low-g-force created by a vortex mixer, rotator, and orbital shaker at various shaking frequencies.³¹ These reports partly explain the relationship between frequency of agitation

and protein aggregate generation; however, no comprehensive study has been carried out because the shaking amplitudes of the apparatuses used in the previous studies were fixed. Thus, the frequency and acceleration were not set independently.

In this study, two therapeutic protein solutions [immunoglobulin G (IgG) and a fusion protein composed of a recombinant IgG Fc domain and the soluble part of the T-cell receptor (CTLA4-Ig)] filled in vials were exposed to three dimensional low-g-force stresses using a transportation test system that controls both acceleration and frequency. Protein aggregates generated by the transportation test system were analyzed using size-exclusion high-performance liquid chromatography (SE-HPLC), quantitative laser diffraction (qLD) and flow imaging microscopy (FIM). A threshold of micron aggregate generation emerged, which was common in three protein formulations. The main route for generating protein aggregates above and below this threshold was discussed. Strategies for mitigating aggregate generation during transportation and discussion of how to more appropriately mimic shipping conditions in laboratories are also presented in this paper.

Materials and methods

Materials

Butyl rubber stoppers, 2-mL borosilicate glass vials (16 × 33 mm), and 2-mL cycloolefin polymer (COP) vials were purchased from Daiwa Special Glass Co., Ltd. (Osaka, Japan). Before use, the rubber stoppers, which were chemically coated with silicone, and the vials were washed three times with ultrapure (18.2 M Ω filtered) water and autoclaved at 121°C for 20 min. Poloxamer 188 (Kolliphor P 188 Bio) was a gift from BASF (Ludwigshafen, Germany). Silicone oil (MED-360) was purchased from Avantor, Inc. (Radnor, PA, USA). Unless otherwise specified, other reagents were purchased from FUJIFILM Wako Pure Chemical Corporation (Osaka, Japan).

Rituximab was used as a model of IgG in vial formulation. IgG (rituximab) was purchased from Zenyaku Kogyo Co., Ltd. (Tokyo, Japan). The IgG solution was dialyzed against 20 mM acetate buffer (pH 5.0) to exchange the buffer from the commercial excipients containing polysorbate 80 and loaded onto an SP Sepharose column (Cytiva, Marlborough, MA, USA), followed by elution with 100 mM phosphate buffer (pH 7.0) containing 400 mM sodium chloride (NaCl). Finally, the protein solution was dialyzed against 20 mM histidine buffer (pH 6.0) or 20 mM histidine buffer (pH 6.0) containing 150 mM NaCl. As a model of relatively less stable proteins, CTLA4-Ig was selected based on a previous study.³³ CTLA4-Ig (Abatacept) was purchased from Ono Pharmaceutical Co., Ltd. (Osaka, Japan). Similar to the procedure used to exchange the buffer of the IgG sample, CTLA4-Ig was diluted with 10 mM phosphate buffer (pH 7.0) and loaded onto a Q Sepharose column (Cytiva), followed by elution with 100 mM phosphate buffer (pH 8.0) containing 400 mM NaCl. Finally, the protein solution was dialyzed against 10 mM phosphate buffer (pH 7.0). CTLA4-Ig in phosphate buffer (pH 7.0) was dialyzed against 10 mM phosphate buffer (pH 5.0) a day before use to minimize oligomerization in this buffer. Poloxamer 188 (P188)-containing buffer was

prepared by a 1:1 dilution of buffer containing P188 at 2-times higher concentration with dialyzed buffer. For example, 1.6% P188-containing phosphate buffer (pH 5.0) was diluted 2-fold with dialyzed phosphate buffer (pH 5.0), and phosphate buffer (pH 5.0) containing 0.8% P188 was obtained. Dialyzed solutions were filtered through 0.22 μm polyethersulfone filters and diluted to 0.5 mg/mL with each buffer before the experiments. A total of 1 mL of protein solution was filled in vials for the experiments.

Stressed stability testing using the transportation test system

Three vials for each condition differ agitated using the transportation test system TRE-200 (ESPEC CORP., Osaka, Japan) (Fig. S1a). When testing at accelerations above 1 G at 10 Hz and 2 G at 20 Hz, modified TRE-200 that produces higher accelerations was used. Vials were fixed in a horizontal orientation to allow the solution to contact the rubber stopper and maximize the air–liquid interface in the vial to simulate the worst-case scenario of degradation. The vials were not packaged and were directly fixed to a polyethylene fixture that was fixed on the top of the transportation test system. Therefore, the vibration was considered to be directly transmitted to the vials. The motion of the transportation test system involved simultaneous triaxial vibration as shown in Figs. S1b and S1c. The acceleration of agitation was adjusted according to the values obtained from the threedimensional wireless vibration recorder MVP-RF8-HC-IDEX (Idex Co., Ltd., Tokyo, Japan), which was fixed on the top of the vial fixture. The acceleration was recorded for each dimensional acceleration of triaxial vibration. In this research, the single-component vectorial acceleration value (G) of the vertical direction of the tri-axial vibration was reported as a representative value. The ratio of x:y:z accelerations was roughly 4:1:5. The g-force of the vertical axis value was the strongest among the three dimensions, which is consistent with transportation by an actual air-ride suspension truck from Hudsonville to Denver (1657 km) (Fig. S2). Three vials were also quiescently placed as control samples.

Frequency and acceleration were set based on actual transportation conditions. For frequency, vibration from 1 to 80 Hz showed higher power than that from 80 to 200 Hz in a power spectral density analysis of truck transportation.³⁰ Additionally, most previous stress stability studies used agitation with a frequency of no more than 50 Hz (3000 rpm).^{31,32} Considering these points, the upper frequency limit of the present study was set at 55 Hz, which is the highest frequency that can be applied with the system. The lower frequency limit was set at 10 Hz, which is the lowest frequency that can be applied by the system. For acceleration of the vertical axis, although the vibration in the vertical axis was 3.2 G at a maximum during actual transportation in an air-ride suspension truck from Hudsonville to Denver, the upper limit of acceleration in the present study was set as 9.7 G to understand the relationship between frequency and acceleration in a wide range. After the agitation or quiescently placed, the samples in three vials were mixed prior to the analysis to obtain sufficient sample solution.

SE-HPLC

SE-HPLC was performed to quantify monomers and nanometer aggregates of samples. For the analysis of IgG,

samples were centrifuged at 25,200 g for 30 min before the analysis to remove insoluble aggregates. Ten microliters of the supernatant was injected three times into an Acquity UPLC H-Class system (Waters, Milford, MA, USA) equipped with an Advance Bio SEC 300 A column (Agilent Technologies, Santa Clara, CA, USA). Separation of the protein species was achieved using a mobile phase consisting of 50 mM sodium phosphate dibasic, 50 mM sodium phosphate monobasic, and 150 mM NaCl. A flow rate of 0.7 mL/min was used. For analysis of CTLA4-Ig, samples were centrifuged at 25,200 g for 30 min before the analysis. Then, 20 mL of the supernatant was injected three times into the Acquity UPLC H-Class system (Waters) equipped with a Tosoh TSK gel G3000SWXL column (Tokyo, Japan). For separation, the mobile phase consisted of 133 mM phosphate buffer and 200 mM potassium chloride, and a flow rate of 0.25 mL/min was used. For both IgG and CTLA4-Ig, UV absorbance detection was performed at 280 nm. Peaks eluted before monomer peak were assigned as nanometer aggregate peaks. Areas under the curves of monomer and nanometer aggregate peaks were analyzed using Empower 3 software (Waters). The relative content of the nanometer aggregates formed after stress tests in percentage was calculated by dividing the peak area of the nanometer aggregates by the total peak areas. The recoveries of the monomer and total soluble proteins (i.e., monomer and nanometer aggregates) were determined by dividing the area of each peak by the areas at the initial time point, as shown in the following equation.

$$R_{\text{monomer}}(t) = \frac{A_{\text{monomer}}(t)}{A_{\text{monomer}}(0)}$$

$$R_{\text{nanometer aggregates}} = \frac{A_{\text{nanometer aggregates}}(t)}{A_{\text{nanometer aggregates}}(0)}$$

$$R_{\text{total}}(t) = \frac{A_{\text{monomer}}(t) + A_{\text{nanometer aggregates}}(t)}{A_{\text{monomer}}(0) + A_{\text{nanometer aggregates}}(0)}$$

Here, $A_{\text{monomer}}(0)$ and $A_{\text{monomer}}(t)$ denote the peak area of the monomer at the initial point and each time point. $A_{\text{nanometer aggregates}}$

(0) and $A_{\text{nanometer aggregates}}(t)$ denote sum of peak areas eluted before monomer peak at the initial point and each time point.

FIM for micron aggregates

A FlowCam 8100™ equipped with an 80 mm flow cell and a 10 × magnification lens (Yokogawa Fluid Imaging Technologies, Inc.,

Scarborough, ME, USA) was used to measure the number concentration of particles ≥ 2 μm (equivalent spherical diameter). Sample analysis volumes were set at 0.15 mL and the flow rate was 0.05 mL/min. Micron aggregates of 2–100 μm in diameter were counted (n = 3). The particle identification settings were 1 μm distance to the nearest neighbor, and the parameter of “close holes” was set as 0. Particle segmentation thresholds of 15.00 were used for both dark and light pixels. VisualSpreadsheet software (Yokogawa Fluid Imaging Technologies) was used for data analysis, and the particles with sizes of no less than 2 μm were included in the analysis.

The samples were analyzed by an Aggregates Sizer with WingSALD Bio software version 3.3 (Shimadzu Corporation, Kyoto, Japan), using the refractive index and density of protein of 1.46 (software's default setting) and 1.32 g/cm³ (based on the density of pure protein),³⁴ respectively. The cut-off level was set as 100, which was effective for reducing the noise that typically results in incorrectly high concentration of particles smaller than 0.3 μm, while the concentration of particles larger than 0.3 μm was unchanged.³⁵ A sample volume of 150 mL was filled into the microcell, and the measurements were performed in accordance with the manufacturer's instructions. The particle diameter was directly estimated by deconvolution analysis of the pattern of light scattered from the sample based on Mie scattering theory.³⁶

Threshold estimation by support vector machine (SVM)

Linear SVM was performed to estimate the threshold of micron aggregate generation. The cost parameter was first optimized for the data of IgG in 20 mM histidine buffer at pH 6.0 with 150 mM NaCl (IgG in HB NaCl) by minimizing the classification error 1000 times in the range of 0.01 to 100 to determine the best-performing SVM model. The cost parameter adds a penalty for each misclassified data point. A lower cost parameter makes a hyperplane with a large margin at the expense of a greater number of misclassifications. Linear SVM was then performed with the optimized cost parameter using input data of the agitating conditions that caused an increase in the number concentration of micron aggregates and the conditions that did not cause such an increase. The obtained hyperplane was applied to IgG in 20 mM histidine buffer at pH 6.0 (IgG in HB) and CTLA4-Ig in 10 mM phosphate buffer (pH 5.0) (CTLA4-Ig in PB), and then the classification accuracy were calculated.

Erosion of adsorbed proteins on the solid surface

Three vials filled with IgG in HB NaCl were agitated using a transportation test system with 50 Hz 9.7 G for 24 h to allow proteins to adsorb on the surface of vials. The solution was removed and the vials were rinsed with ultrapure (18.2 MΩ filtered) water at least seven times to remove unadsorbed proteins. The histidine–NaCl buffer was then added to the vials. The three vials for each condition were quiescently placed or agitated using a transportation test system with 50 Hz 6.3 G or 50 Hz 3.5 G. The samples in three vials were mixed prior to the analysis to obtain sufficient sample solution, and the number concentration of micron aggregates was measured by FIM (n = 3).

Erosion of adsorbed silicone oil on the solid surface

Two grams of silicone oil in toluene (3 w/w%) was added to vials and incubated overnight at room temperature to coat the solid surface of the vials with this oil, as described previously.¹⁹ After removing the solution, the vials were incubated at 40°C until they

dried. The silicone oil-coated vials were used as a surrogate of vials with

Table 1
Parameters evaluated for threshold of micron aggregate generation.

Parameter	Condition
Viscosity	0.9 cP*: IgG in HB NaCl 8.9 cP: IgG in HB NaCl containing 54% sucrose 18.9 cP: IgG in HB NaCl containing 66% sucrose
Protein concentration	0.5*, 5, and 10 mg/mL
Vial size	16 ± 33 mm* and 18 ± 40 mm
Quantity of solution	1 mL* and 2 mL
Vial orientation	Horizontal* and vertical
Temperature	Room temperature* and 5°C

* Conditions written in Stressed stability testing using the transportation test system.

protein film layers. Histidine–NaCl buffer was then added to the vials. The three vials for each condition were quiescently placed or agitated using a transportation test system with 50 Hz 6.3 G or 50 Hz 3.5 G. The samples in three vials were mixed prior to the analysis to obtain sufficient sample solution, and the number concentration of eroded silicone oil droplets was counted by FIM (n = 3).

High-speed video analysis

The motion of protein solutions in vials was recorded using a high-speed camera (Fastcam Mini AX200; Photron, Tokyo, Japan) at a rate of 10,000 frames per second with a light source, while the vials were agitated at various combinations of frequency and acceleration: 10 Hz 2 G, 10 Hz 0.7 G, 50 Hz 6.3 G, and 50 Hz 3.5 G. For the agitating conditions shown in Table 1, the videos were taken at a rate of 1000 frames per second with a light source. The recorded videos were then analyzed using Photron FASTCAM Viewer 4 software.

Stressed stability testing with different parameters

Viscosity, protein concentration, vial size, quantity of solution, vial orientation, and temperature were varied as shown in Table 1 to examine the impact of the parameters on micron aggregate generation. High-viscosity solutions were prepared by adding sucrose to IgG in HB NaCl. Viscosities of IgG in HB NaCl, IgG in HB NaCl containing 54% sucrose (IgG54suc), and IgG in HB NaCl containing 66% sucrose (IgG66suc) measured by a VROC Initium viscometer (RheoSense, San Ramon, CA) at 23°C were 0.9, 8.9, and 18.9 cP, respectively. One parameter was varied at a time, and other parameters were kept the same to conditions written in Stressed Stability Testing Using a Transportation Test System. Three vials filled with protein solutions for each condition were agitated for 24 h at 50 Hz 3.5 G, 4.1 G, and 4.5 G using TRE-200. The samples in three vials were mixed prior to the analysis to obtain sufficient sample solution. The number concentration of micron aggregates after agitation was measured by FIM (n = 3), and the impacts of these parameters on the threshold of micron aggregate generation were evaluated.

Results

Stress testing of IgG in HB NaCl using the transportation test system

The effect of vibration during truck transportation on protein aggregate generation was investigated by tri-axial vibration, which is similar to the actual conditions experienced during transportation with a truck.³⁷ The focus here was on IgG in HB NaCl for up to 24 h, with various combinations of frequency and acceleration using a transportation test system.

Fig. 1a and d show the number concentration of micron aggregates (≥ 2 μm) formed after 24 h agitation and the observed rate coefficient, k_{obs} , defined as the slope of the linear plot of monomer recovery against monomer area of the initial. The data used for the determination of k_{obs} is shown in Fig. S3. The regression line for estimating k_{obs} was obtained over the time range where a linear relationship was established (squared value of correlation coefficient ≥ 0.85). Agitating conditions that increased the number concentration of micron aggregates from the initial value by $> 1332/\text{mL}$ are shown in red. Conditions that conferred no apparent increase in the number concentration of micron aggregates ($< 1332/\text{mL}$) are shown in blue. The value of $1332/\text{mL}$ represents the mean plus three standard deviations (SD) of the measured number concentration of micron aggregates (≥ 2 μm) at the initial time point (Table S1). Marked increases in the number concentration of micron aggregates were observed when acceleration exceeded a specific value across the tested frequency range (hereinafter referred to as the threshold of micron aggregate generation). Although the result at 30 Hz 2.7 G included a variability and showed a remarkably higher number concentration of micron aggregates compared to the other conditions, the increase of number concentration of micron aggregates from the initial value was observed consistently (Fig. S4). k_{obs} was less sensitive than measuring the number concentration of micron aggregates; however, k_{obs} also increased as acceleration increased. No increase in a relative content of nanometer aggregates from the initial time point was observed under all conditions examined (Table S2). The number concentration of submicrometer aggregates (0.3 to 2 μm) was quantified by qLD.^{35,36} The submicrometer aggregates were detected only above the threshold of micron aggregate generation, and the number concentration of submicrometer aggregates showed similar trend to the results obtained by FIM (Fig. S5).

Stress testing of IgG in HB and CTLA4-Ig in phosphate buffer

The results raised the question of whether the threshold for IgG in HB NaCl can be observed in other proteins and formulations. Thus, IgG in HB was used in the agitation-stress test to investigate the effect of formulation on aggregate formation (Fig. 1b, c and e). This formulation was selected because IgG would have higher colloidal stability in the absence of NaCl in the buffer, which could reduce the number

concentration of aggregates.^{38,39} Conditions that yielded an increase in the number concentration of micron aggregates and conditions that did not cause such an increase were distinguished based on the mean plus 3 SD value at the initial time point (i.e., $767/\text{mL}$; Table S1). Compared with IgG in HB NaCl, as expected, lower number concentration of micron aggregates were generated, and the monomer loss rate was slower for IgG in HB. Notably, conditions of frequency and acceleration, which caused a significant increase in the number concentration of micron aggregates, were similar to those of IgG in HB NaCl.

The stability of another protein against the agitation stresses was investigated using CTLA4-Ig in 10 mM phosphate buffer (pH 5.0) (CTLA4-Ig in PB). The pH of the formulation was close to the experimental isoelectric point of CTLA4-Ig (pH 5.1), in accordance with our previous research,¹⁵ and it was thus expected to be less stable than the IgG formulations. CTLA4-Ig in PB showed a higher number concentration of micron aggregates following stress for 6 h and higher k_{obs} than the IgG formulations (Fig. 2a, b, and S6). Conditions that yielded an increase in the number concentration of micron aggregates and conditions that did not cause such an increase were distinguished based on the mean plus 3 SD value at the initial time point ($20,673/\text{mL}$; Table S1). Although CTLA4-Ig in PB showed lower stability than IgG formulations, the threshold of micron aggregate generation was similar to those for IgG in HB NaCl and IgG in HB.

Different behaviors were observed for a relative content of nanometer aggregates between IgG and CTLA4-Ig samples. The increase in the relative content of nanometer aggregates was observed under quiescent conditions only in CTLA4-Ig in PB. In addition, agitation

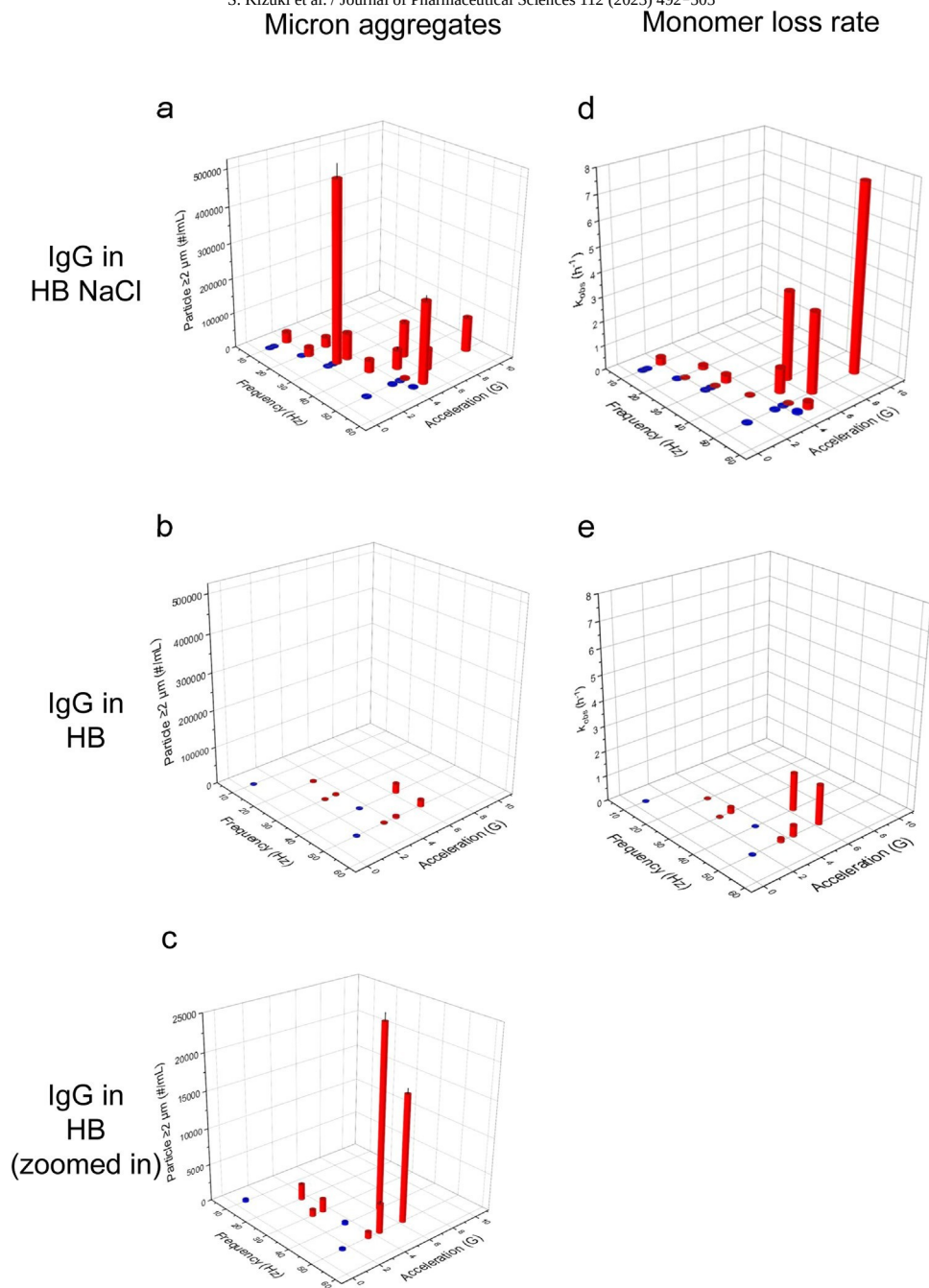


Fig. 1. Micron aggregates and monomer loss rate of IgG in HB NaCl and IgG in HB agitated for 24 h by various combinations of frequency and acceleration using a transportation test system. The number concentration of micron aggregates (≥ 2 mm) in IgG in HB NaCl (a), in IgG in HB (b), and zoomed in view of (b) (c). Error bars represent the SD of triplicate measurements of FIM. The monomer loss rate, k_{obs} , in IgG in HB NaCl (d) and IgG in HB (e). Red: the number concentration of micron aggregates was higher than 3 SD above the mean count value at the initial time point. Blue: both the number concentration of micron aggregates and the relative content of nanometer aggregates were not increased.

accelerated nanometer aggregate generation in CTLA4-Ig in PB (Fig. 2c and d).

Threshold of increasing micron aggregates in IgG in HB NaCl, IgG in HB, and CTLA4-Ig in PB

As shown in Figs. 1a, c and 2a, an increase in the number concentration of micron aggregates occurred over a specific range of frequencies and accelerations. A boundary of micron aggregate generation was drawn by SVM, which is a classification method that uses supervised machine learning.⁴⁰ The central concept of SVM involves identifying

a hyperplane that is used as a decision surface to maximize the gap between two classes.

FIM results obtained for IgG in HB NaCl after agitation stresses were used for the training. A cost parameter was first optimized to determine the best-performing SVM. Linear SVM was conducted 1000 times using cost parameters from 0.1 to 100. The classification error was minimal when the cost parameter was 67.1418. The obtained hyperplane was applied to IgG in HB and CTLA4-Ig in PB, and then the classification accuracy was calculated. Interestingly, all the data for IgG in HB and CTLA4-Ig in PB were

classified into two classes with high accuracy (79.0%) (Fig. 3b and c).

P188 were evaluated to determine whether oligomerization was interface-mediated. As shown in

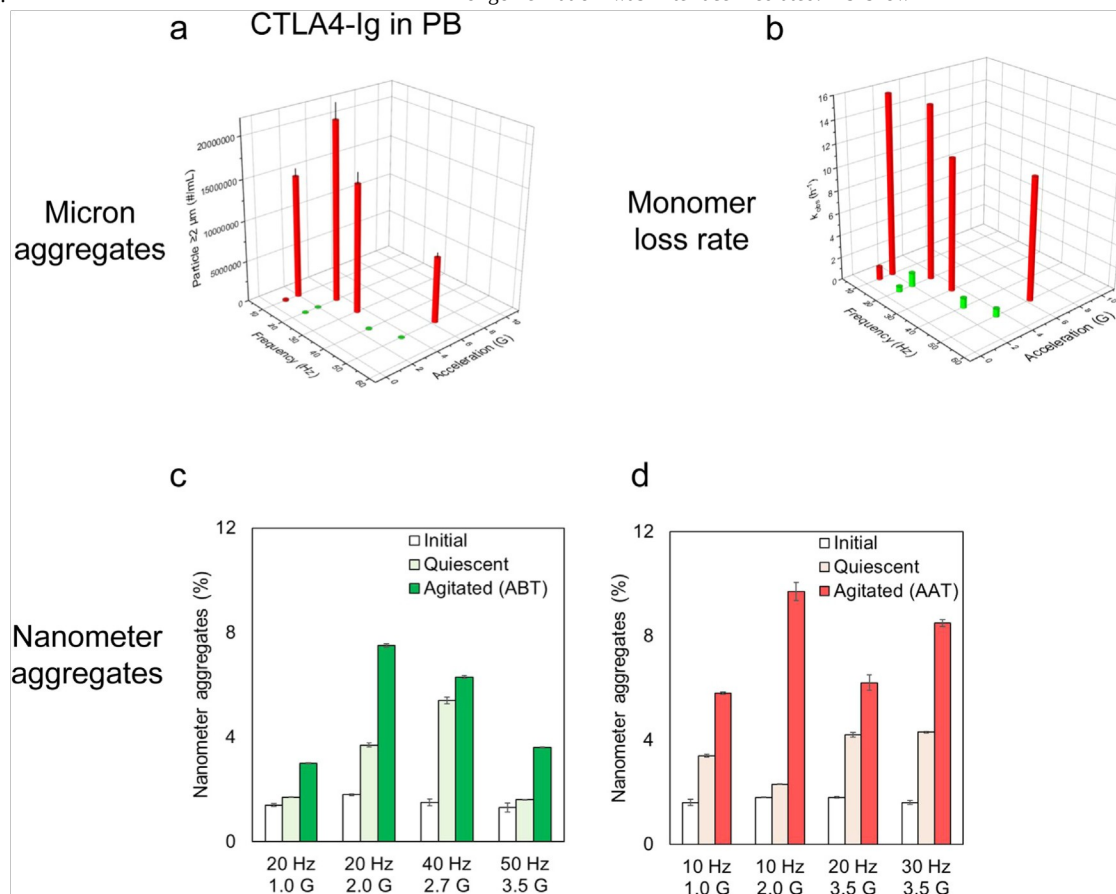


Fig. 2. Micron aggregates and monomer loss rate of CTLA4-Ig in PB agitated by various combinations of frequency and acceleration using a transportation test system. (a) The number concentration of micron aggregates ($\geq 2 \mu\text{m}$) after agitation for 6 h. Error bars represent the SD of triplicate evaluation of FIM. (b) The monomer loss rate, k_{obs} . Red: the number concentration of micron aggregates was higher than 3 SD above the mean count value at the initial time point. Green: the number concentration of micron aggregates was not increased, but spontaneous oligomerization was accelerated. Nanometer aggregates in CTLA4-Ig in PB stressed with agitation below the threshold of micron aggregate generation

(ABT) (c) or agitation above the threshold of micron aggregate generation (AAT) (d).

Participation of solid–liquid and air–liquid interfaces in aggregate generation above and below the threshold

To investigate the contribution of the solid–liquid interface (i.e., the inner surface of vials) and air–liquid interface to formation of aggregates by low g-force stresses, 0.008%, 0.08%, and 0.8% (w/v) P188, which is known to prevent interface-mediated aggregates effectively, were added to CTLA4-Ig in PB. The distribution range of P188 includes 0.8% in the marketed formulation [Orencia Syringe for S.C. Injection (Ono Pharmaceutical Co., Ltd.)], which is above the critical micelle concentration (CMC) of P188 [0.105% (w/v)].⁴¹ Agitation above the threshold of micron aggregate generation (AAT) with 0.08% and 0.8% P188 did not increase the number concentration of micron aggregates of CTLA4-Ig (Fig. 4a). Although the number concentration of micron aggregates increased in 0.008% P188-containing solution, this number concentration was much lower than that in the P188-free formulation. This result implies that the participation of interface in AAT induced micron aggregate generation.

The relative contents of nanometer aggregates of CTLA4-Ig in PB containing 0.008%, 0.08%, and 0.8%

Fig. 4b and c, the relative contents of nanometer aggregates were increased by both AAT and agitation below the threshold of micron aggregate generation (ABT) even in the presence of 0.8% P188.

To further investigate the impact of the air–liquid interface on nanometer aggregates, CTLA4-Ig in PB with 0.8% P188 was agitated below the threshold of micron aggregate generation (50 Hz 3.5 G) for 6 h with or without an air–liquid interface. Although agitation increased relative contents of nanometer aggregates, no difference was observed in the contents with or without the air–liquid interface (Fig. 4d).

Next, the impact of the solid–liquid interface on nanometer aggregates was investigated using CTLA4-Ig in PB with 0.1% P188 filled in a COP or borosilicate glass vial. We previously demonstrated that protein adsorption differs between these two materials.¹⁵ The vials were agitated below the threshold of micron aggregate generation (50 Hz 3.5 G) for 6 h, and generated nanometer aggregates were measured by SE-HPLC. As shown in Fig. 4e, although the agitation increased the relative contents of nanometer aggregates, there was no difference in the contents of nanometer aggregates between COP and borosilicate vials.

Investigating the participation of solid–liquid interface mediated aggregate formation by AAT

To confirm the contribution of the solid-liquid interface on micron aggregate generation by AAT, we compared monomer loss rates and the number concentration of micron aggregates after agitation in two types of vial made from borosilicate glass or COP. The experiments were performed only for IgG in HB NaCl because no increase of nanometer aggregates was observed. The monomer loss rate constant k_{obs} of IgG in HB NaCl in borosilicate glass vials was observed to be 1.7 times higher than that of the same sample in COP vials (k_{obs} was 1.93 for COP and 3.28 for borosilicate glass). Moreover,

Fig. 3. Threshold of micron aggregate generation. (a) The threshold was obtained by SVM using data of IgG in HB NaCl as training data. The hyperplane obtained by (a) was depicted in IgG in HB (b), and CTLA4-Ig in PB (c). Red: the number concentration of micron aggregates was higher than 3 SD above the mean count value at the initial time point. Blue: No increase in the number concentration of micron aggregates.

more micron aggregates were generated in borosilicate glass vials than in COP vials after agitation for 3 and 6 h (Figs. S7a and S7b). These results suggested that micron aggregates increased through the solid-liquid interface by the stress of AAT. The participation of the air-liquid interface is discussed later.

Impact of agitation on eroding protein films on the solid-liquid interface

The participation of the solid-liquid interface under AAT stress conditions was further clarified by evaluating the number concentration of eroded protein films from the solid-liquid interface by AAT and ABT using IgG in HB NaCl. Initially, the solution was agitated above the threshold, and then the vial was washed with ultrapure water to remove unbound proteins. Subsequently, histidine-NaCl buffer was added, followed by AAT or ABT. After applying stress, the number concentration of micron aggregates was measured using FIM.

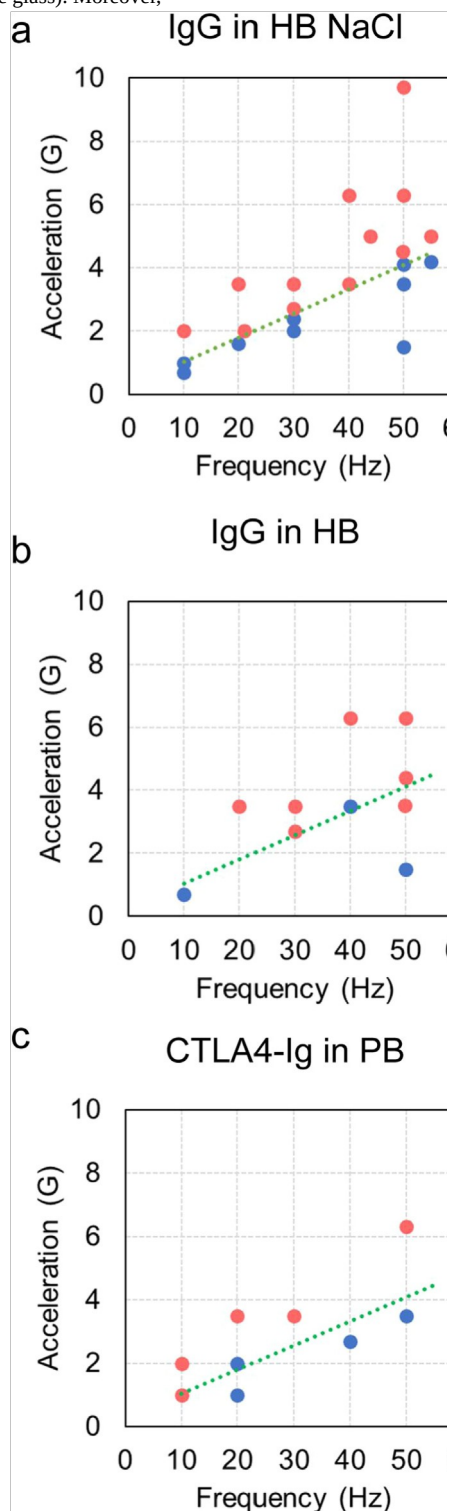
No increase in micron aggregates was observed after undergoing quiescence or stress with ABT (Fig. 5a). In contrast, micron aggregates increased with time after the stress test by AAT.

The efficiency of the erosion by AAT was then evaluated using silicone oil-lubricated vials. Silicone oil was used as a model of a layer adsorbed to the solid surface. In this experiment, silicone oil was lubricated onto the vial wall before applying stress. Then, the erosion of the silicone oil was examined under quiescent, ABT, and AAT conditions. The number concentration of silicone oil droplets increased by AAT was approximately 18 times greater than that by ABT (Fig. 5b), indicating that the silicone oil layer was eroded efficiently by AAT.

To examine whether an air-liquid interface participates in erosion of the silicone oil layer by AAT, a silicone oil eroding experiment was conducted with AAT without a headspace (air-liquid interface). The number concentration of silicone oil droplets was not increased by AAT (Fig. 5b). This indicated that the air-liquid interface plays an important role in erosion from the solid-liquid interface.

High-speed image analysis of fluids in vials stressed by AAT and ABT

Next, videos of the vials with AAT and ABT at a rate of 10,000 frames per second at 10 Hz and 50 Hz were recorded using a highspeed camera. The images of AAT showed an irregular pattern and disruption of the waves (Figs. 6a and 6c, Movies S1 and S3), whereas relatively regular waves were observed for vials of ABT (Figs. 6b and 6d, Movies S2 and S4). The vigorous waves by AAT generated higher compression/dilation ratio in air-liquid interface than ABT. Note that no visible cavitation was recorded in both agitated vials, indicating that no cavitation effect was responsible for erosion by agitation.



Factors influencing on the threshold of micron aggregate generation

Thresholds of micron aggregate generation were examined under different conditions. Fill volume, vial orientation, vial size, viscosity, temperature, and protein concentration were varied, as shown in [Table 1](#).

The total number concentrations of micron aggregates ($\geq 2 \mu\text{m}$) at the initial state and after the agitation for 24 h at 50 Hz 3.5 G, 50 Hz 4.1 G, and 50 Hz 4.5 G are shown in [Fig. 7a](#). Agitation conditions that increased the number concentration of micron aggregates from the initial state are shown in pink. An increase of micron aggregates was determined when the number concentration of particles was higher than the mean plus 3 SD of number concentration of particles ($\geq 2 \mu\text{m}$) at the initial time point. Larger fill volume and vertical vial orientation resulted in micron aggregate generation at lower acceleration than the lower fill volume and horizontal vial orientation. Higher g-force was required to generate micron aggregates when using a larger vial (18 $\pm$ 40 mm). Other parameters did not influence on the threshold of micron aggregate generation. High-speed camera images of vials during agitation at 3.5 G, 4.1 G, and 4.5 G of acceleration at 50 Hz showed irregular protruding waves only under the conditions that increased micron aggregate ([Fig. S8](#)).

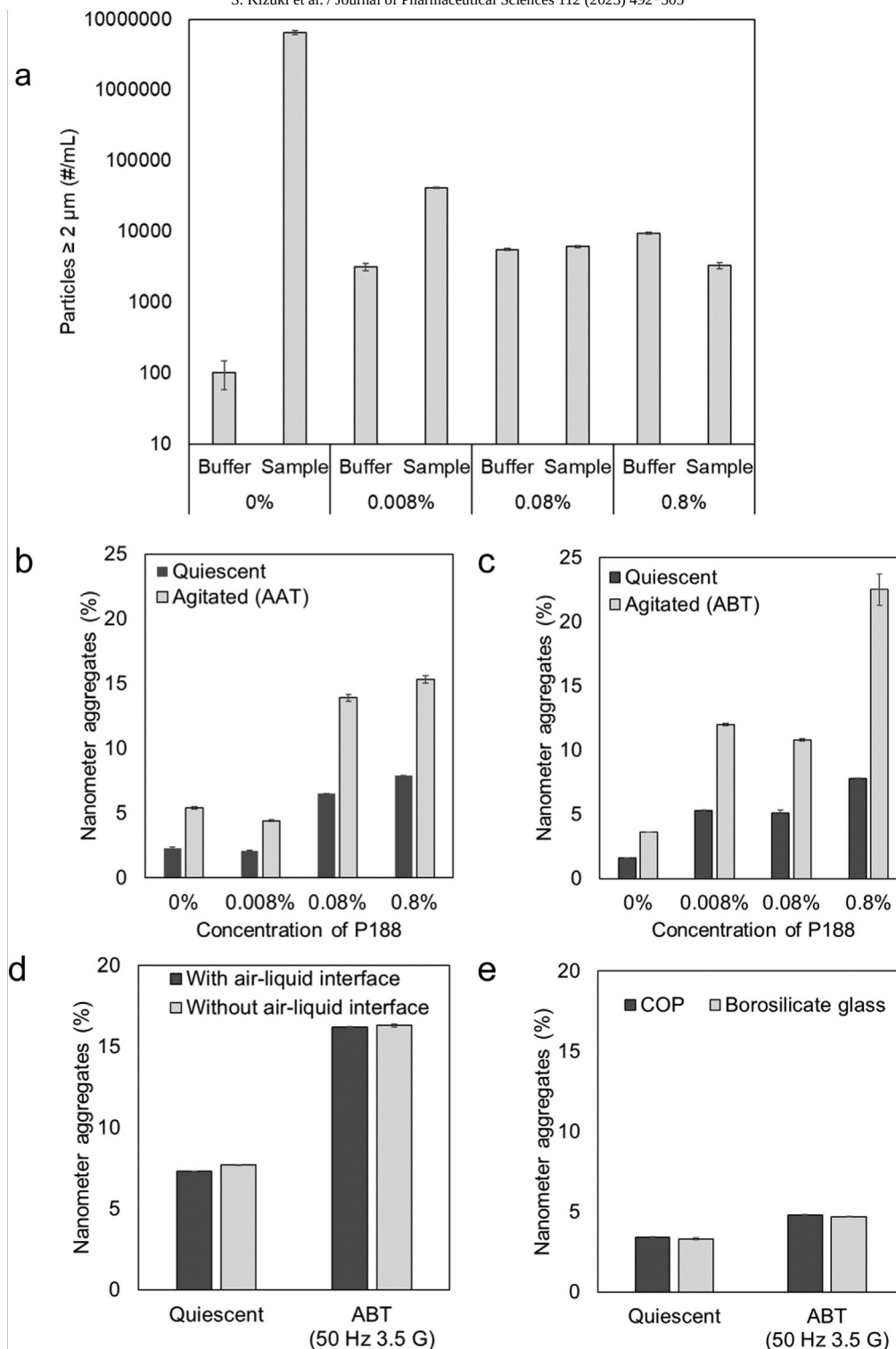


Fig. 4. Impact of P188 on nanometer aggregate generation of CTLA4-Ig after stressing with AAT or ABT for 6 h. (a) The number concentration of micron aggregates in samples of CTLA4-Ig in PB containing 0%, 0.008%, 0.08%, and 0.8% P188 stressed by AAT. Error bars represent the SD of triplicate measurements of FIM. Nanometer aggregates in CTLA4-Ig in PB and CTLA4-Ig in PB containing 0%, 0.008%, 0.08%, and 0.8% P188 stressed with AAT (50 Hz 6.3 G) (b) or ABT (50 Hz 3.5 G) (c). Nanometer aggregates in CTLA4-Ig in PB containing 0.1% P188 stressed with agitation at 50 Hz 3.5 G with or without air-liquid interface (d). Nanometer aggregates in CTLA4-Ig in PB containing 0.1% P188 stressed with agitation at 50 Hz 3.5 G in COP and borosilicate glass vials (e). Error bars from (b) to (e) represent the SD of triplicate measurements of SE-HPLC.

Discussion

The importance of low-g-force stresses in damaging protein solutions during transportation has been documented.²⁷ However, the relationship between aggregate generation and agitation parameters such as frequency and acceleration has not been characterized in detail. The

present study focused on low g-force stresses generated by a transportation test system.

In the experiment that agitations at various combinations of frequency and acceleration were applied to two pharmaceutical

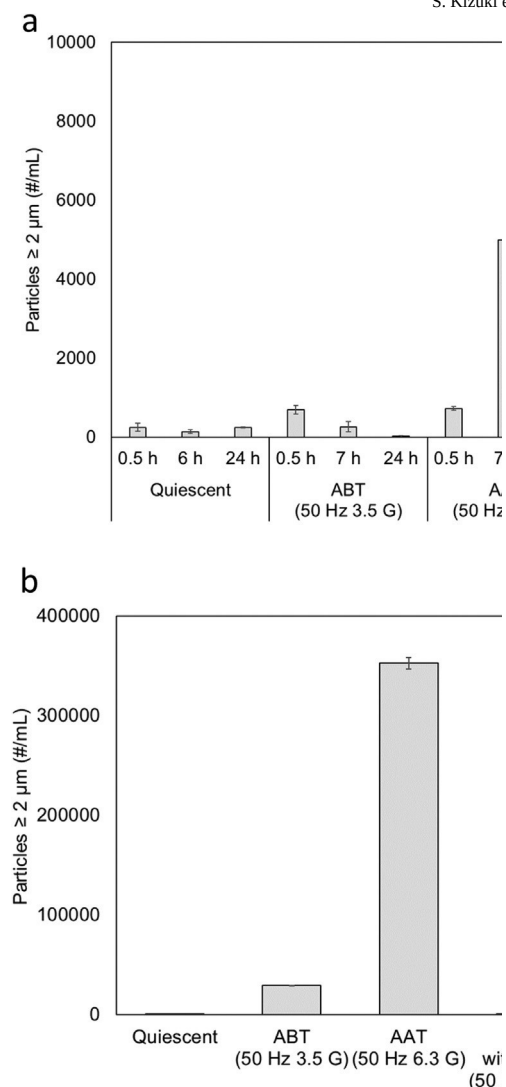


Fig. 5. Histidine NaCl buffer was added to the vial after adsorbing IgG or lubricating silicone oil to the solid surface. The vials were quiescently stored or vibrated for evaluating erosion from the solid surface. (a) The number concentration of micron aggregates ($\geq 2 \mu\text{m}$) was determined after 0.5, 6, and 24 h under quiescent conditions. The number concentration of micron aggregates eroded from the solid-liquid interface by ABT (50 Hz 3.5 G) and AAT (50 Hz 6.3 G) for 0.5, 7, and 24 h. (b) The number concentration of particles eroded from a silicone oil-coated vial under quiescent conditions, stress test with ABT (50 Hz 3.5 G), AAT (50 Hz 6.3 G), and AAT in a filled vial to remove the air-liquid interface. Error bars represent the SD of triplicate measurements of SE-HPLC measurement.

proteins (IgG and CTLA4-Ig), interface-mediated and bulk-mediated aggregate generation were observed. AAT generated micron aggregates mainly via interface-mediated routes, whereas agitation, including ABT, accelerates spontaneous oligomerization in bulk solutions. Interestingly, the thresholds of micron aggregate generation that dissects the whole region into two regions based on the behavior of micron aggregate generation were common among the three protein formulations without surfactant, suggesting that characteristics of protein would not have a large impact on the threshold of micron aggregate generation by low-g-force stress.

A proposed mechanism of micron aggregate generation by AAT

Mechanical stresses are recognized to induce interface-mediated aggregation, and non-ionic surfactants that prevent proteins from interfacial damage can mitigate the generation of protein aggregates.^{15,16,17,42} As consistent with the previous studies, an increase in the number concentration of micron aggregates by AAT was suppressed efficiently by the addition of 0.8% P188. A recent study reported that the adsorption of monoclonal antibody (mAb) to the air-liquid interface after rupture of the interface was completely inhibited when mAb coexisted with 0.0001% polysorbate 80, which was lower than the CMC of polysorbate 80 (0.0015%).⁴³ Although surfactant and proteins used in our study were different from them used in the previous study, proteins would not adsorb to the air-liquid interface in the presence of 0.8% P188 even during AAT which rupture the air-liquid interface continuously. There is a possibility that micron aggregate generation by AAT was bulk-mediated, and P188 prevents it by interacting with hydrophobic patches of proteins.⁴² However, nanometer aggregates were increased by both AAT and ABT with or without P188 in the present study, implying that protein-protein interaction was not weakened by P188. Thus, the results of experiments using P188 indicated that micron aggregates were generated by AAT mainly through the interface-mediated route.

Results obtained in this study suggest that the solid-liquid interface plays an important role in micron aggregate generation by AAT. The monomer loss rate and the number concentration of micron aggregates generated by AAT were influenced by the difference in vial materials (COP and borosilicate glass). This indicated that the solid-liquid interface participates in micron aggregate generation by AAT because the material of the solid surface has been shown to influence on micron aggregates when the aggregate generation route is solid-liquid interface-mediated.^{17,44} We further investigated the mechanism of solid-liquid interface-mediated micron aggregate generation by AAT from the step of erosion of protein films adsorbed to the solid surface. In erosion experiments, protein or silicone oil-coated vials were stressed with AAT and ABT with protein-free buffer. A higher number concentration of micron aggregates or silicone oil droplets were generated by AAT than ABT. Thus, the high efficiency of erosion by AAT would be one of the reasons why AAT generate higher number concentration of micron aggregates. Gerhardt et al. showed that micron particles consisting of protein and silicone oil increased upon agitation stress in a siliconized syringe.¹⁹ They suggested that the movement of three interfaces (silicone oil coated solid surface–water–air) by agitation caused capillary forces that pulled and ruptured the layer of silicone oil and gelled protein aggregates at the solid-liquid interface. In our study, three interfaces (solid-liquid–air) moved faster in a wider area with AAT than with ABT (Movies S1–S4). The importance of the three interfaces was supported by the result that the erosion of silicone oil was prevented by the removal of the air-liquid interface from the vial.

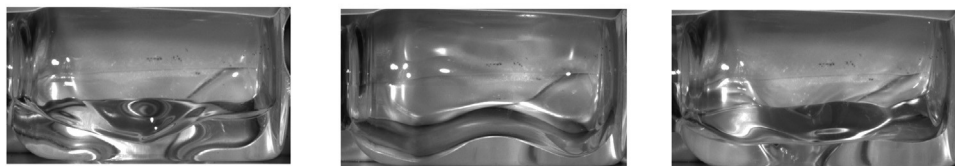
The air-liquid interface is also reported to participate in micron aggregate generation. The compression/dilation ratio of interfacial area is an important factor of micron aggregate generation.²⁵ Bee et al. suggested that the critical compression/dilation ratio

is five, and air-liquid interface-mediated aggregates markedly increased when the ratio is higher than five.²⁶ AAT generated a higher compression/dilation ratio than ABT because AAT generates more vigorous and protruding waves than ABT. A cycle number of compression/dilation is also an important factor of micron aggregate generation at the air-liquid interface. Ghazvini et al. showed mAb samples subjected to 750–1000 compression/dilation cycles contained a higher number concentration of micron aggregates in bulk solution, whereas mAb samples subjected to 50 cycles showed no increase of aggregates in bulk solution, despite aggregates present at air-liquid interface.¹⁶ Taking the agitation condition (frequency and duration) of AAT into account, the cycle number of compression/dilation must be >1000. Since the waves generated by AAT meets the criteria for both compression/dilation ratio and cycle number, AAT may induce the air-liquid interface-mediated aggregate generation.

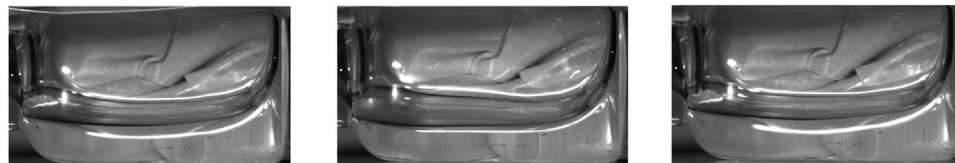
oligomerize. The addition of 0.008%, 0.08%, and 0.8% P188 to the formulation did not prevent oligomerization. Because 0.8% P188 is a higher proportion than that of CMC (0.105%), most of the solid-liquid and air-liquid interfaces must be covered by surfactant; thus, the observed acceleration in oligomerization would occur via a bulk-mediated route. This was supported by the two experiments. Nanometer aggregates of CTLA4-Ig in PB with 0.8% P188 were not suppressed by removal of headspace (air-liquid interface). Vial material did not influence on the increase of nanometer aggregates, although the surface materials have an impact on the amount of protein adsorbed on the solid phase. Although the detailed mechanism was unclear, we postulate that the agitation enhances the mixing efficiency, and protein molecules collide more frequently.

When the buffer includes P188, relative content of nanometer aggregates was increased compared with the result without surfactant. A previous report showed

(a) AAT(10 Hz and 2.0 G)



(b) ABT (10 Hz and 0.7 G)



(c) AAT (50 Hz and 6.3 G)



(d) ABT (50 Hz and 3.5 G)



Fig. 6. High-speed camera images of vials containing 1 mL of 0.5 mg/mL IgG in HB NaCl vibrated at 10 Hz with AAT (2.0 G) (a) and ABT (0.7 G) (b). Images of the same vials vibrated at 50 Hz with AAT (6.3 G) (c) and ABT (3.5 G) (d).

A proposed mechanism of accelerating spontaneous oligomerization by agitation

Both AAT and ABT accelerated the spontaneous oligomerization of CTLA4-Ig in PB. In contrast, an increase in the relative content of nanometer aggregates was not observed in IgG in HB and IgG in HB NaCl. These results indicate that agitation accelerates oligomerization only when proteins spontaneously

nanometer aggregates increased when polysorbates were added in the buffer, and the authors in the study proposed the possibility of interaction of polysorbate with the protein.⁴⁵ Similarly, P188 might interact with CTLA4-Ig and caused the increase of nanometer aggregates.

Implication of the threshold of micron aggregate generation

As shown in Fig. 3, the threshold of micron aggregate generation for IgG in HB NaCl showed a linear relationship between frequency and acceleration. Considering that the threshold obtained from IgG in HB NaCl could be applied

to the other formulations (IgG in HB and CTLA4-Ig in PB), we hypothesized that fluid dynamics of the solution would be a key to understanding the threshold of micron aggregate generation. On the basis of this assumption, videos of solutions under AAT and ABT taken by a high-speed camera were compared between AAT and ABT. Images of the fluid stressed by AAT showed irregular and disrupted waves, whereas the waves under ABT conditions moved repetitively. This surface disruption would be related to the threshold of micron aggregate generation. Hashimoto et al. previously reported that increase of the acceleration of vertical oscillation destabilized the waves of a solution in a cylinder. The disruption of wave surfaces and the breakup of drops were observed when acceleration exceeded a specific point.⁴⁶ The previous study also showed the threshold of surface-breaking was a linear relationship between acceleration and frequency over the range of 10 to 60 Hz, which was similar to the linear threshold of micron aggregate generation in this study that revealed higher acceleration was necessary to induce micron aggregates as frequency became higher. The amplitude of the agitation decreases at higher frequency if the acceleration is constant. A lower amplitude of agitation can induce lower-amplitude waves that do not rupture. Thus, higher acceleration is needed at a higher frequency to increase the amplitude and to break the wave.

Although there is no direct evidence from this study, there is another possibility that the micron aggregate generation might be limited by diffusion/adsorption to the interface, and the limitation might be a cause of the upward threshold. The solution might reach a different steady state at higher frequencies by molecular diffusion transport limitation.

Factors influencing threshold of micron aggregate generation

The threshold of micron aggregate generation was examined with various formulation-related parameters (Fig. 7). Parameters were set with a consideration of real-world situations; for example, protein concentration was set as high as the marketed formulation of rituximab (10 mg/mL), and viscosity was set up to 20 cP because solutions with viscosity above 20 cP are considered to be difficult to inject.⁴⁷

Protein concentration, viscosity, and temperature did not influence the threshold of micron aggregate generation. Larger fill volume and vertical vial orientation made the threshold lower, and larger vial size made the threshold higher. These results may be explained from the viewpoint of height and dimension at the base area ratio of the solution (H/D ratio). As the H/D ratio increased, a surface rupture occurred by vibration with smaller amplitude.⁴⁶ In other words, surface rupture can occur easily at high H/D ratio. In fact, irregular protruding waves were found at lower acceleration when the H/D ratio was higher (larger fill volume and vertical vial orientation). This means that the threshold of micron aggregate generation can be influenced by factors which have impact on the fluid dynamics. The threshold of micron aggregate generation of different container types, such as pre-filled syringes would be different from that of vial formulation, and needs to be evaluated in future work.

Suggestion for evaluating stability against mechanical stresses during a shipping test in the laboratory

One of the primary purposes of this study was to establish a more appropriate shipping test in the laboratory. As described above, Fleischman et al. successfully simulated micron aggregate generation during transportation using a random vibration test in accordance with ASTM D4169 with a transportation simulator. The only method that predicted real-time shipment was the random vibration test. Thus, the transportation simulator's random vibration test is important for predicting the generation of micron aggregates in actual shipping.

In the random vibration test, a time-compressed setting may shorten the test period by increasing the acceleration of vibration.⁴⁸ Although this acceleration is useful for cargo, applying this approach to protein solutions must be carefully considered according to our finding of the threshold of micron aggregate generation. Assuming the case where the vibration level during actual transportation is below the threshold of micron aggregates, the time-compressed setting may exceed the threshold of micron aggregate generation, leading to an incorrect interpretation of the aggregate generation. Avoiding time compression is a conservative way of simulating aggregate generation properly. In fact, a pharmaceutical company usually did not compress the simulation time because accelerated testing did not match the data from actual shipping.⁴⁹ Thus, the use of time-compressed settings requires prior evaluation to compare real-time shipping and the time-compressed settings.

Forced degradation studies using horizontal and orbital shakers are widely used to evaluate stability against agitation stress.^{50,51,52} The acceleration values of the horizontal^{53,54,55} and orbital^{31,45} shaking stresses in the previous studies were estimated based on the information about rotation speed and amplitude of the orbital shaker, and were below 2 G, which were considered low-g-force stresses.²⁷ The stress conditions in the previous study that generated and did not generate insoluble aggregates were plotted in Fig. S9 in red and blue, respectively. A line extrapolated from the threshold obtained in this study passed on two kinds of data points at which an increase of micron aggregates was observed or not observed. This indicates that the threshold obtained with three-dimensional shaking in this study could be valid for the agitation stresses by orbital shakers.

Suggestion for mitigating aggregate generation by transportation stresses

The addition of nonionic surfactants, such as polysorbate and P188, is effective and the current industry standard to prevent interface-mediated aggregate formation. However, polysorbates are degraded by oxidation and hydrolysis, and some degradation products may disrupt protein stability.^{56,57} Although P188 is more chemically stable than polysorbates, a recent study suggested that P188 may undergo oxidative degradation under particular stress conditions.⁵⁸ Our findings suggest that the risk of micron aggregate generation due to the interfacial stress can be mitigated if the vibration levels are suppressed to below the threshold of micron aggregate generation by

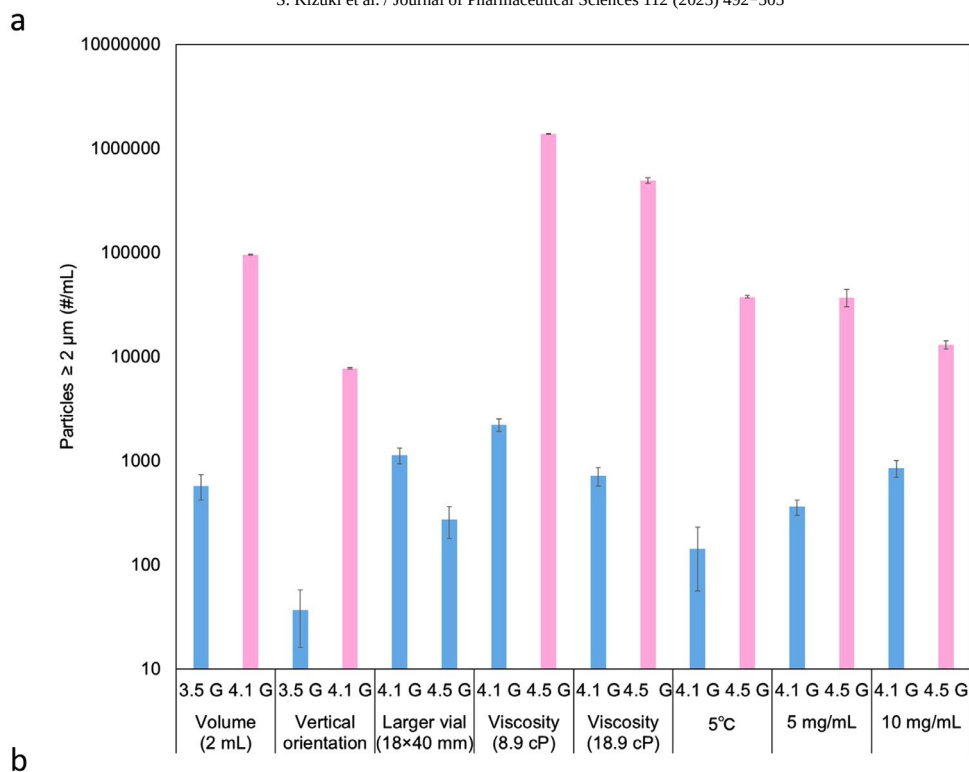
optimization of the packaging and shipping processes (e.g., air-ride suspension truck). Optimization of the solid material of container and closure to reduce protein adsorption/erosion would also be an effective approach because solid-liquid interface participated in micron aggregate generation by AAT. These approaches may make it possible to lower the concentration of or even to omit surfactants in protein formulations. Formulations with lower concentration of surfactants could be beneficial because surfactants may increase nanometer aggregates as shown in this study and the previous study⁴⁵. Also, micron aggregates related to degradants of surfactants do not need to be cared when the surfactants are not included in formulations.

The acceleration of spontaneous oligomerization that occurred in bulk solution also needs to be considered. Since the spontaneous oligomerization was observed only in CTLA4-Ig in PB which was relatively unstable, the spontaneous oligomerization can be prevented by appropriate protein sequence and formulation developments, although non-ionic surfactant was not effective.

Conclusions

Significant micron aggregate generations were observed when agitation stresses in a certain range of acceleration and frequency were applied to vials containing protein solutions. The threshold of micron aggregate generation was common in three formulations employed in this study. This would be because fluid dynamics is more important for micron aggregate generation than characteristics of protein. Actually, parameters which can affect fluid dynamics such as fill volume and vial size had impact on the threshold. Although further study is necessary, the threshold related to fluid dynamics could be applied for other formulations.

In this work, it was shown that micron aggregate generation by AAT occurred mainly via interface-mediated route and can be prevented effectively by the addition of surfactant. The increase of nanometer aggregates was accelerated even with the surfactant by agitation under the formulation condition that spontaneous protein oligomerization occurred without agitation. We propose two types of mitigation strategies for alleviating both micron and nanometer aggregate generation from transportation stress: reduction of agitation stress not to exceed threshold (e.g. optimizing packaging and transportation) and optimization of protein sequence and formulation.



*: Conditions written in *Stressed stability testing using the transportation test system*.

Fig. 7. Threshold of micron aggregate generation with different parameters related to vial formulation. The number concentration of micron aggregates ($\geq 2 \mu\text{m}$) was shown (a). Increase of micron aggregates was examined based on the values of mean plus 3 SD value of unstressed samples (5 mg/mL: 871 #/mL, 10 mg/mL: 999 #/mL, IgG54suc: 3567 #/mL, and IgG66suc: 2191 #/mL. For the other parameters, 1332 #/mL was used.). Pink: the number concentration of micron aggregates was increased. Gray: the number concentration of micron aggregates was not increased. Error bars represent the SD of triplicate measurements of FIM measurement. Results were tabulated (b). Y: the number concentration of micron aggregates was increased from the initial, N: the number concentration of micron aggregates was not increased from the initial.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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regarding fluid dynamics of the waves in vials. Finally, the authors thank Edanz (<https://jp.edanz.com/ac>) for editing a draft of this manuscript. This work was partly supported by AMED (20ak0101074h1504 and 21ae0121013j0001 to S.U.).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.xphs.2022.09.022](https://doi.org/10.1016/j.xphs.2022.09.022).

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